Unsupervised Domain Adaptation Based on Progressive Transfer for Ship Detection: From Optical to SAR Images

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***Abstract*— In recent years, synthetic aperture radar (SAR) ship detection methods based on convolutional neural networks (CNNs) have attracted wide attention in remote sensing fields. However, these methods require a large number of labeled SAR images to train the network, where labeling for SAR images is more expensive and time-consuming than that for optical images. To address the problem of lacking labeled SAR images, in this article, we proposed an unsupervised domain adaptation (UDA) framework based on progressive transfer for SAR ship detection by transferring knowledge from the optical domain to the SAR domain. Due to the prominent difference between the optical and SAR images, our approach progressively transfers knowledge from three levels: pixel level, feature level, and prediction level. At the pixel level, considering the difference in imaging mechanism, we propose a special data augmentation method for ship targets and build the generator with skip connection based on generative adversarial networks (GANs) to generate transition domain, which can reduce the appearance discrepancy between the optical and SAR images. At the feature level, the detector is trained to learn the domain-invariant features with adversarial alignment. At the prediction level, we further use the predicted pseudo-labels obtained by the feature-aligned detector to learn more discriminative features of the SAR images directly and propose the robust self-training (RST) method to reduce the influence of noisy pseudo-labels on detector training. Specially, RST is formulated as a loss minimization problem for object detection. The experimental results based on the domain adaptation from optical dataset to SAR dataset demonstrate that our approach achieves superior SAR ship detection performance with unlabeled SAR images.**

***Index Terms*— Generative adversarial networks (GANs), optical to synthetic aperture radar (SAR), robust self-training (RST), SAR, ship detection, unsupervised domain adaptation (UDA).**

I. Introduction

AS AN active microwave imaging sensor, synthetic aperture radar (SAR) has been extensively used in the military and civilian fields with the capability of monitoring targets under all-day and all-weather conditions. In recent years, ship detection in SAR images, which is a great branch of SAR image interpretation, has attracted wide attention because of the important value in ocean monitoring, military intelligence acquisition, and so on [1]–[4].

In traditional ways, constant false alarm rate (CFAR), which compares the gray value of the pixels with a certain adaptive threshold through a sliding window, is the most widely used SAR target detection method [5]. However, it will cause a performance drop when the scene is complex in practice. It is difficult for CFAR to choose a statistical distribution model suitable for nonuniform strong clutter background in complex scenes. Recently, the methods based on convolutional neural networks (CNNs) have achieved remarkable success in object detection fields by the way of data-driven. Among these, faster R-CNN [6], a two-stage detector, has achieved superior performance using a region proposal network (RPN) to produce target proposals and then processing target proposals with the two branches of classification and regression heads. Due to its excellent performance and flexibility of expanding to other tasks, faster R-CNN has been widely used in many fields. In terms of SAR ship detection, Li *et al.* [2] used a feature fusion strategy and proposed an improved faster R-CNN. Based on faster R-CNN, Cui *et al.* [3] proposed a dense attention pyramid network for multiscale ship detection. Zhao *et al.* [4] introduced a new attention mechanism and combined it with the feature pyramid in the faster R-CNN to detect multiscale ships.

Although SAR ship detection methods based on CNNs have achieved remarkable performance, they still require large amounts of labeled data to train the detector, which is expensive and time-consuming, especially for SAR images with complex scenes and clutters. Generally, SAR images are challenging to interpret for ordinary people, and annotating SAR images requires professionally trained experts. Therefore, obtaining large-scale labeled training SAR images becomes an obstacle to train a deep detector. Relatively speaking, optical images are easier to be obtained and labeled. Instead of training a model with labeled data directly, unsupervised domain adaptation (UDA), which aims to transfer knowledge from the fully labeled source domain (such as optical domain) to the unlabeled target domain (such as SAR domain), provides an excellent way to solve the problem of lacking labeled data.

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In computer vision fields, UDA for recognition and semantic segmentation has been extensively studied, roughly divided into three categories.

1) Pixel-level methods based on generative adversarial networks (GANs). Most of these methods aim to translate the source images to the target images [7]–[9] to alleviate domain discrepancy using image-to-image translation GANs, such as CycleGAN [10].
2) Feature-level methods based on feature alignment. These methods aim to learn the domain-invariant feature representation to align the distributions of features between the source domain and the target domain based on the maximum mean discrepancy (MMD) [11]–[14], or gradient reversal layer (GRL) [15]–[17].
3) Pseudo-labeling methods based on self-training. Noticing the similarity in problem settings between UDA and semisupervised learning (SSL), the methods based on self-training are attempted in UDA for recognition and segmentation [18]–[20] and are proved as effective UDA methods [21].

In more recent researches on SAR recognition, Rostami *et al.* [22] exploited knowledge transfer from the optical domain to the SAR domain. Huang *et al.* [23] explored transfer learning with MMD from multiple domains (including optical domain) to the SAR.

Although the UDA method has received much attention in recognition and segmentation, UDA for object detection is still in its infancy, mainly focusing on feature alignment. Chen *et al.* [24] aligned the distribution of features between the source and the target domains from image and instance feature level through adversarial learning. Following [24], Saito *et al.* [25] trained a cross-domain detector based on strong local alignment and weak global alignment. Shen *et al.* [26] and He and Zhang [27] proposed multiadversarial learning to learn domain-invariant features at low-, middle-, and high-level feature spaces. Chen *et al.* [28] further proposed a similarity-reweighted adversarial module to reweight the interpolated features. Besides that, the methods based on pixel level are also explored [29]–[32]. Arruda *et al.* [29] and Shan *et al.* [30] translated the source images to the target images by CycleGAN and trained the detector with generated images. Shi *et al.* [31] generated target images via CycleGAN to alleviate the problem of limited SAR images and learned useful features of SAR images by feature alignment and discriminative learning, which can transfer knowledge from one SAR domain to another SAR domain. Kim *et al.* [32] generated multiple translated domains by GANs with color preservation and reconstruction constraint and learned multidomain-invariant representation with adversarial learning.

The above methods provide a feasible insight to transfer knowledge to unlabeled target domain from the source domain whose labeled data are easy and inexpensive to obtain. Following this line of work, in this article, we aim to tackle the challenges of detecting ships in the unlabeled SAR domain by transferring knowledge from the labeled optical domain. To the best of our knowledge, there is no UDA research yet on SAR ship detection. Due to the prominent difference between the optical and SAR images, current approaches for the detection of optical images cannot be applied for SAR images directly. The main problems remaining to be solved can be described as follows.

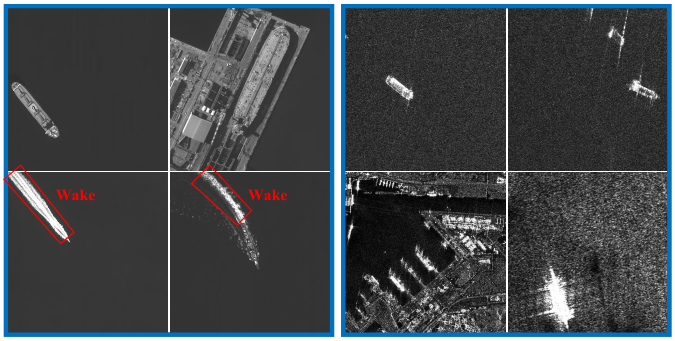


Fig. 1. Some examples of ship images as (a) optical images and (b) SAR images.

1) Due to the different imaging mechanisms, as shown in Fig. 1, SAR and optical images significantly differ in appearance. Unlike optical images, SAR images mainly include texture, structure, and shape information without gray, color, and brightness information [33]. Especially for ship detection tasks, the wakes of the ship in the optical images are more obvious and intense, which makes cross-domain ship detection more difficult.
2) The features of SAR and optical images have a large difference. For example, the SAR features are sensitive to the changes in view angles, parameters, etc., and optical features vice versa. In computer vision fields, feature-aligned methods can reduce the feature discrepancy. However, most of the existing works have focused on UDA from optical to optical images. The difference lies only in the image style but their imaging mechanism is the same. Therefore, they can transfer at low-, middle-, and high-level shared feature spaces, which are not suitable for optical-to-SAR transfer. As pointed in [34], the distributions of optical and SAR data are probably too different from each other to transfer in low layers. How to transfer from optical domain to SAR domain remains to be explored.
3) Due to the massive difference between the SAR domain and the optical domain, the method based on feature alignment will lose some unique and valuable features of the SAR domain when moving the feature distribution closer to the labeled optical domain. The pseudo-labeling methods, which can train the model with the images of the target domain directly, provide an excellent way to solve such a problem. However, since the pseudo-labels are not completely correct, wrong labels will hurt detector training. Therefore, how to train a robust detector to reduce the influence of noisy labels on detector training is a challenge.

Due to the above challenges, it is not easy to transfer knowledge from the optical domain to the SAR domain immediately. In this article, an UDA based on progressive transfer for ship detection is proposed, which can gradually transfer the

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knowledge between the optical domain and the SAR domain from pixel level, feature level, and prediction level. At the pixel level, we aim to translate the optical images to transition images similar to SAR images using data augmentation and GANs. First, to alleviate the wakes of ships and complex land background on ship target generation, we extract ship targets and synthesize them with the pure ocean background to enhance the quality and diversity of ship target generation. In the training process, the original and synthesized images participate in the training together, which can be seen as a particular data augmentation method for ship targets. Besides that, considering the particularity of SAR images that mainly include texture, structure, and shape information without gray, color, and brightness information, we build a generator with skip connection to preserve low-level texture, structure, and shape information. At the feature level, considering the higher similarity at high-level semantic feature space between optical and SAR domains, the domain-invariant features are learned at high-level feature space using adversarial learning to align feature distributions. At the prediction level, we further use the predicted pseudo-labels obtained by the feature-aligned detector (treated as a teacher model) to learn more discriminative knowledge of the SAR images directly and proposed the robust self-training (RST) method to reduce the influence of noisy pseudo-labels on detector training. Our RST trains a robust detector from the following aspects: 1) jointly training the detector with accurately labeled transition domain data; 2) selecting high-quality pseudo-labels; and 3) label smooth. Specially, we treat the unknown labels of SAR images as latent variables (pseudo-labels), and RST is formulated as a loss minimization problem for object detection, which can be solved by iteratively producing pseudo-labels and retraining the detector. Besides that, the proposed RST can also be used for semisupervised object detection tasks. More specifically, the main contributions of the proposed method are as follows.

1) *UDA Framework Based on Progressive Transfer for Ship Detection Is Proposed:* To the best of our knowledge, it is the first work for the unsupervised cross-domain ship detection from optical to SAR remote sensing images.
2) *Novel Pixel-Level-Based Method Is Proposed to Reduce the Appearance Discrepancy Between the Source and Target Domains:* Considering the different imaging mechanisms, we propose a data augmentation method for ship targets to enhance the quality and diversity of ship target generation and build the generator with skip connection to generate high-quality transition images.
3) *RST Is Proposed to Learn More Discriminative Features and Alleviate the Influence of Noisy Pseudo-Labels on Detector Training:* Specially, RST is formulated as a loss minimization problem for object detection. Besides that, the proposed RST can also be used for semisupervised object detection tasks.

In the remainder of this article, Section II introduces the details of the proposed approach based on progressive transfer for ship detection. Section III presents the experiments and analysis based on the domain adaptation from the optical dataset to SAR dataset. Finally, the conclusion is drawn in Section IV.

II. Proposed Domain Adaptation Method

The technical details of the proposed UDA framework based on progressive transfer for SAR ship detection will be introduced in this section. The overall architecture of the proposed method for SAR ship detection is shown in Fig. 2, which progressively transfers knowledge from three levels: pixel level, feature level, and prediction level. The proposed method is based on faster R-CNN. Because faster R-CNN is a simple and typical two-stage detector that can be flexibly expanded to other tasks. In addition, it is also the most popular unsupervised domain adaptative detector in computer vision fields [35], and only very few methods choose SSD [36] as the detection model. In this way, we can more easily compare with the existing UDA methods for object detection in computer vision fields. But the idea of our UDA for ship detection is also general to many other deep object detection models, such as SSD, RefineDet [37], and FPN [38].

Formally, in the task of cross-domain ship detection, we have access to a fully labeled source domain (optical domain) $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, where x_i^s and y_i^s denote the optical remote sensing image and the corresponding ship annotations, respectively. An unlabeled target domain (SAR domain) $\mathcal{D}_t = \{(x_j^t)\}_{j=1}^{N_t}$ is given, where x_t^s represents the unlabeled SAR remote sensing image. The target domain is divided into two parts: one is the training data and the other is test data. The goal of our method is to learn a cross-domain detector that can perform well on the test data of unlabeled SAR domain by transferring the knowledge from a fully labeled optical domain to the training data of unlabeled SAR domain. In the following, the details of each level will be introduced.

A. Pixel Level: Optical-to SAR Image Generation

Due to different imaging mechanisms, SAR and optical images significantly differ in image appearance. To mitigate such difference, we follow the previous pixel-level adaptation method based on GAN and aim to generate a transition domain that cannot be distinguished from SAR domain by the discriminator, while the locations of the ship targets remain unchanged from the source domain. Therefore, we enable our detector to learn on labeled transition domain while still generalizing to SAR domain.

1) Data Augmentation for Ship Targets: When GAN generates images from optical to SAR, each image pixel, whether a background pixel or a target pixel, is treated equally, which may generate some unsatisfactory ship data. For example, the wakes of the ship in some optical images are so intense, and some land backgrounds in some optical images are so complex that the ship targets are not focused on by the GAN. To alleviate such a problem, as shown in Fig. 3, we synthesize the ship target data with a pure ocean background to reduce the impact from ship wakes and complex land backgrounds on ship data generation. In the synthesis process, we make certain disturbances to the position and pose angle of the

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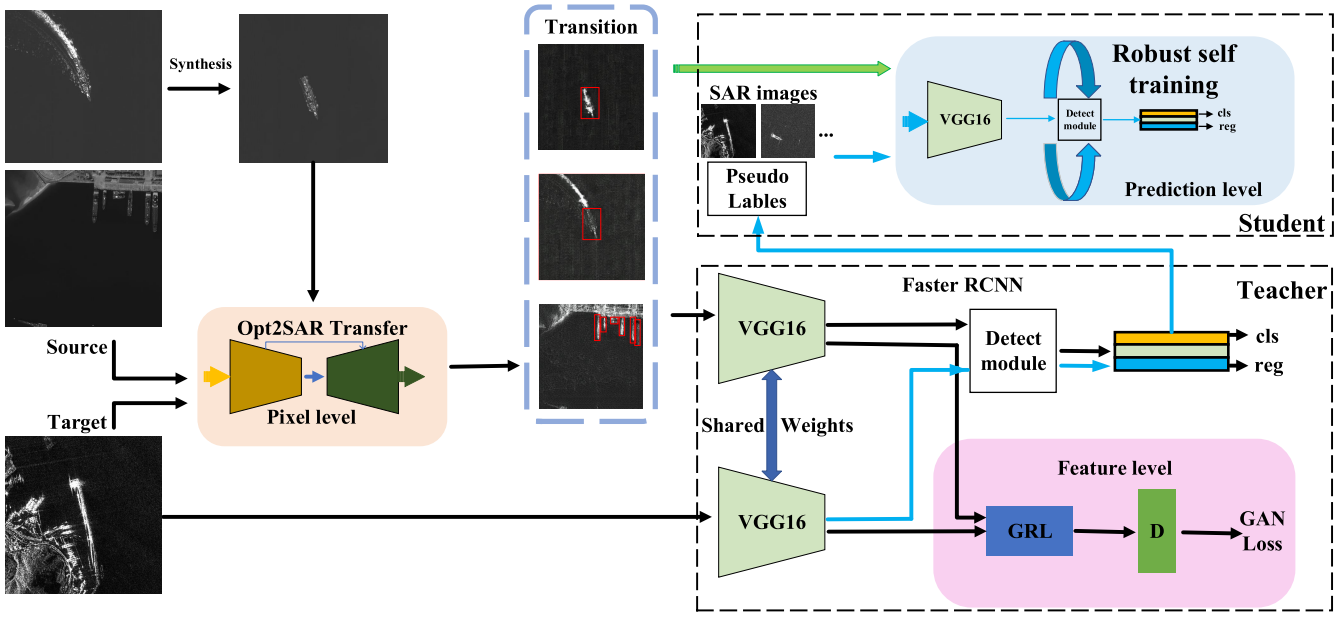


Fig. 2. Architecture of the unsupervised cross-domain adaptation framework based on progressive transfer for SAR ship detection. At the pixel level, Opt2SAR transfer transfers the source and synthesis images to transition images. At the feature level, the parameters of VGG16 are shared between two domains and GAN loss is used to learn the domain-invariant feature representation. The blue lines represent the flow of pseudo-labels' generation. At the prediction level, we treat the detector obtained in the previous stage as a teacher model and generate the pseudo-labels by testing on the unlabeled training target SAR images.

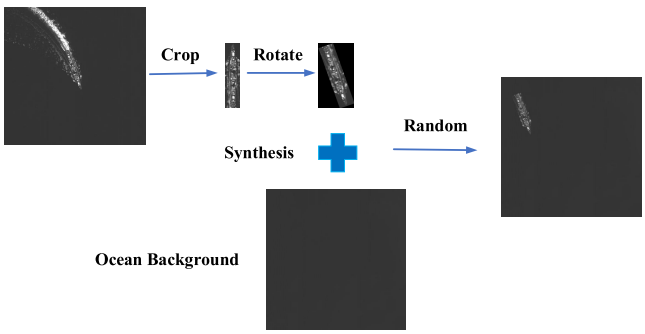


Fig. 3. Process of synthesizing ship target and background.

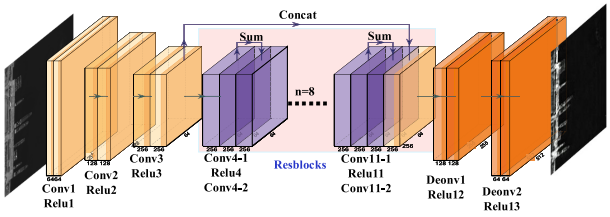


Fig. 4. Architecture of the generator with skip connection.

ship targets to enhance the diversity. In the training phase of optical-to-SAR GAN, the original image and the synthesized image participate in the training to generate the transition domain. In the subsequent training process of feature level and prediction level, the images generated by the original image and the synthetic image are trained together, which can be seen as a particular data augmentation method for ship targets.

2) Optical-to-SAR GAN: Then, we will introduce an optical-to-SAR GAN composed of the generator and discriminator to achieve the generation of transition domain. The generator $G_{O\to S}$ is trained to produce SAR images from optical images to fool the discriminator D_{SAR}. Conversely, the discriminator D_{SAR} is trained to classify whether the image comes from real SAR data or generated SAR data (transition domain). This adversarial process is trained with the GAN loss function

$$\begin{aligned}\mathcal{L}_{\mathrm{GAN}}(G_{O\to S}, D_{\mathrm{SAR}}, X_{\mathrm{opt}}, X_{\mathrm{SAR}}) &= \mathrm{E}_{x_{\mathrm{SAR}}\sim X_{\mathrm{SAR}}}[\log D_{\mathrm{SAR}}(x_{\mathrm{SAR}})] \\ &\quad + \mathrm{E}_{x_{\mathrm{opt}}\sim X_{\mathrm{opt}}}[\log(1 - D_{\mathrm{SAR}}(G_{O\to S}(x_{\mathrm{opt}})))]. \end{aligned} \tag{1}$$

However, the huge difference between the optical and SAR images makes the generator difficult to generate satisfying images. Unlike optical images, SAR images mainly include texture, structure, and shape information, and these characters are mainly included in the low-level feature space of CNNs. To ensure the generator producing convincing SAR images, we further build the generator with a skip connection. The architecture of the generator is improved on the basis of [39]. As shown in Fig. 4, our generator contains three stride-2 convolutions, eight residual blocks with sum-operation skip connection, and two deconvolutions with stride 1/2. Specially, we add the concat-operation skip connection between the third layer and third-to-last layer to preserve low-level texture, structure, and shape information.

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Although the GAN loss can make the generator generate data drawn from X_{SAR}, there is no way to guarantee the generator preserves the overall arrangement of images and the location and pose of ship targets. Following the mainstream of image-to-image translation methods, we use the cycle consistency constraint [10], [40], [41], another generator $G_{S\to O}$, and another discriminator D_{opt} on our optical-to-SAR GAN. The generator and discriminator are trained with the same GAN loss $\mathcal{L}_{\mathrm{GAN}}(G_{S\to O}, D_{\mathrm{opt}}, X_{\mathrm{opt}}, X_{\mathrm{SAR}})$. The cycle consistency constraint makes $G_{S\to O}(G_{O\to S}(x_{\mathrm{opt}})) \approx x_{\mathrm{opt}}$ and $G_{O\to S}(G_{S\to O}(x_{\mathrm{SAR}})) \approx x_{\mathrm{SAR}}$. The cycle consistency loss is described as

$$\begin{aligned}&\mathcal{L}_{\mathrm{cycle}}(G_{O\to S}, G_{S\to O}, X_{\mathrm{opt}}, X_{\mathrm{SAR}})\\&\quad=\mathrm{E}_{x_{\mathrm{opt}}\sim X_{\mathrm{opt}}}[||G_{S\to O}(G_{O\to S}(x_{\mathrm{opt}})-x_{\mathrm{opt}}||_1]\\&\qquad+\mathrm{E}_{x_{\mathrm{SAR}}\sim X_{\mathrm{SAR}}}[||G_{O\to S}(G_{S\to O}(x_{\mathrm{SAR}})-x_{\mathrm{SAR}}||_1]. \quad (2)\end{aligned}$$

The final loss is as follows:

$$\begin{aligned}&\mathcal{L}_{\text{opt-to-SAR-GAN}}\\&\quad=\alpha\mathcal{L}_{\mathrm{GAN}}(G_{\mathrm{opt}\to\mathrm{SAR}}, D_{\mathrm{SAR}}, X_{\mathrm{opt}}, X_{\mathrm{SAR}})\\&\qquad+\mathcal{L}_{\mathrm{GAN}}(G_{\mathrm{SAR}\to\mathrm{opt}}, D_{\mathrm{opt}}, X_{\mathrm{opt}}, X_{\mathrm{SAR}})\\&\qquad+\beta\mathcal{L}_{\mathrm{cycle}}(G_{\mathrm{opt}\to\mathrm{SAR}}, G_{\mathrm{opt}\to\mathrm{SAR}}, X_{\mathrm{opt}}, X_{\mathrm{SAR}}) \quad (3)\end{aligned}$$

where α and β are the balance parameters.

B. Feature Level: Domain-Invariant Feature Representation

In this stage, we focus on reducing the feature difference and transferring knowledge from the labeled transition domain to the unlabeled SAR domain at the feature level. Since the labels can only be acquired by the transition domain, the detector is expected to extract the common features of the two domains, that is, the domain-invariant features, to guide the learning of the SAR data domain. To this end, we aim to train a detector that extracts domain-invariant features to align feature distributions between the generated images and SAR images. Noticing that the huge difference between the two domains, we only align distributions at high-level feature space (last convolutions in the feature extractor, such as VGG16 in Fig. 2) instead of low and middle levels. We use domain adversarial learning to achieve this process. Given an image x_i, the corresponding high-level feature $F(x_i)$ extracted by feature extractor F, and a domain classifier D, the goal of D is to classify whether the feature $F(x_i)$ comes from the target domain or generated transition domain, and the goal of $F(x_i)$ is to fool D. The objective of domain adversarial learning can be written as

$$\begin{aligned}&\mathcal{L}_{\mathrm{adv}}(F, D, X_{\mathrm{trans}}, X_{\mathrm{SAR}})\\&\quad=\mathrm{E}_{x_{\mathrm{trans}}\sim X_{\mathrm{trans}}}[\log D(F(x_{\mathrm{trans}}))]\\&\qquad+\mathrm{E}_{x_{\mathrm{SAR}}\sim X_{\mathrm{SAR}}}[\log(1-D(F(x_{\mathrm{SAR}})))]. \quad (4)\end{aligned}$$

The domain classifier D is trained to maximize this objective, and the feature extractor F is trained to minimize this objective. The GRL [15] is used to connect them. After this adversarial process, F tends to learn the domain-invariant feature representation.

The detection loss of the detection task can be divided into classification loss and location regression loss, and it is as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{det}}&=\mathcal{L}_{\mathrm{cls}}+\mathcal{L}_{\mathrm{loc}}\\&=\mathcal{L}_{\mathrm{cls}}^{\mathrm{rpn}}+\mathcal{L}_{\mathrm{loc}}^{\mathrm{rpn}}+\mathcal{L}_{\mathrm{cls}}^{\mathrm{roi}}+\mathcal{L}_{\mathrm{loc}}^{\mathrm{roi}}. \quad (5)\end{aligned}$$

The overall objective loss of the feature level can be written as

$$\mathcal{L}=\mathcal{L}_{\mathrm{det}}+\lambda\mathcal{L}_{\mathrm{adv}} \quad (6)$$

where λ is the balance parameter.

C. Prediction Level: Robust Self-Training

As mentioned in Section II, domain-invariant feature representation will ignore the domain-specific features. In this stage, we treat the detector obtained in the previous stage as a teacher model and generate the pseudo-labels by testing on the unlabeled training target SAR images. Then, supervised training with pseudo-labels, called self-training, can be applied to learn more discriminative representations of the SAR images. However, since the pseudo-labels are not completely correct, the wrong pseudo-labels will hurt detector training. We further proposed the RST method to reduce the influence of wrong labels on detector training. Our RST trains a robust detector from the following aspects.

1) Jointly training the detector with accurately labeled transition domain data. Although the images of the transition domain still have a gap with real SAR images, the transition domain's labels are completely correct. Jointly training the detector with accurately labeled transition domain data can alleviate the impact of the detector being guided by wrong labels.
2) Selecting high-quality pseudo-labels. Most self-training methods for semisupervised detection only select high-confident proposals as the pseudo-labels and ignore the overall quality of the pseudo-labels in an image [42]–[44]. As shown in Fig. 5, the images in which correct pseudo-labels are more, such as Image A, will benefit the training of the detector. We call such images as easy images. The images in which wrong pseudo-labels are more, such as Image B, will hurt the training of the detector. We call such images as difficult images. For these difficult images, when setting a high threshold (e.g., 0.85), some ships will be treated as backgrounds. When setting a low threshold (e.g., 0.6), some backgrounds will be treated as ships. Such images should not be selected to train the detector. To alleviate such a problem, our method selects easy images not only at the proposal level but also at the image level. We first select proposals at the proposal level for each image by setting a confidence threshold. Then we calculate the average confidence score of all the selected proposals inside an image and sort all the images according to their average confidence scores from high to low. We further select top p percent images to participate in training. The details of this aspect will be introduced later.

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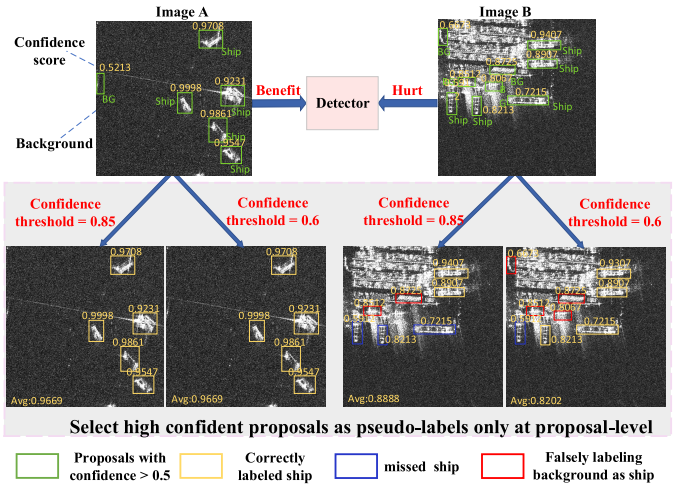


Fig. 5. Selecting proposals as pseudo-labels at the proposal level for two SAR images. For Image A, the correct pseudo-labels are more whether setting a high confidence threshold or a low confidence threshold. For Image B, wrong pseudo-labels are more. When setting a high threshold, some ships will be treated as backgrounds; when setting a low threshold, some backgrounds will be treated as ships.

3) Label smooth. The label smooth makes the detector not wholly believe in the accuracy of the labels. Therefore, label smooth is introduced to RST to enhance the fault tolerance ability.

Specially, RST is formulated as a loss minimization problem for detection and can be solved by iteratively producing pseudo-labels and retraining the detector. Next, we will introduce our formulation process for self-training and RST of UDA detection in detail.

1) Formulation for Self-Training of UDA Detection: If the detection annotations of source/transition and target domain are both accessible, the easiest method to transfer knowledge is fine-tuning on both the domains. However, in UDA ship detection settings, the SAR domain labels cannot be accessible. Another way is self-training, which treats the labels of target images as latent variables that can be automatically learned. For ship detector with classification output and location output, we present the UDA loss of the faster RCNN for the RPN head for simplicity. The UDA detection is formulated as the following objective:

$$\begin{aligned}\min_{w} \mathcal{L}_{\mathrm{ST}}(w) = &-\sum_{s=1}^{S}\sum_{b=1}^{N}\Big[y_{s,b}^{\top}\log(\mathrm{cls}_b(\mathbf{w}, i_s)) \\ &\qquad + \mathrm{smooth}_{L_1}\left(\mathrm{loc}_b(\mathbf{w}, i_s) - v_{s,b}^{\top}\right)\Big] \\ &-\sum_{t=1}^{T}\sum_{b=1}^{B}\Big[\hat{y}_{t,b}^{\top}\log(\mathrm{cls}_b(\mathbf{w}, i_t)) \\ &\qquad + \mathrm{smooth}_{L_1}\left(\mathrm{loc}_b(\mathbf{w}, i_s) - \hat{v}_{s,b}^{\top}\right)\Big] \\ \text{s.t. } &\hat{y}_{t,b} \in \{0, 1\} \quad \forall t, b \end{aligned} \tag{7}$$

where i_s represents the input image, and s is the index of the source/transition domain. $y_{s,b}^{\top}$ and $v_{s,b}^{\top}$ denote the ground-truth category and location labels for the bth proposal bounding boxes in i_s, respectively, and $\mathbf{w}$ denotes the detector weights. $\mathrm{cls}_b(\mathbf{w}, i_s)$ and $\mathrm{loc}_b(\mathbf{w}, i_s)$ are the category output with probability and location output. i_t, $\mathrm{cls}_b(\mathbf{w}, i_t)$ and $\mathrm{loc}_b(\mathbf{w}, i_t)$ are similar for the target domain. Unlike the source/transition domain, $\hat{y}$ in the target domain denotes the pseudo category labels of proposals, which can be automatically learned by the detector, and assigning it as 1 leads to treating this pseudo-label as target ground truth and 0 as background otherwise. Other areas of this image are treated as background.

2) Formulation for RST of UDA Detection: The pseudo-labels are not completely correct and it is difficult to optimize the pseudo-labels on the unsupervised target domain. The self-training method generally follows the "easy to hard" strategy to select the most confident pseudo-labels (easy pseudo-labels). The detector is updated with these most confident pseudo-labels and then regenerates the difficult pseudo-labels with less confidence. We call this process self-training with proposal-level selection. However, for the detection task, it should not only consider proposal-level selection. As mentioned in the previous section, the pseudo-label's quality of image level is also essential. We hope to select the images with more correct knowledge (easy images) to train the detector and then regenerate pseudo-labels for those with more wrong knowledge (difficult images). The revised self-training that consists of proposal-level selection and image-level selection is formulated as follows:

$$\begin{aligned}\min_{\mathbf{w},\hat{y}} \mathcal{L}_{\mathrm{ST}}(\mathbf{w}, \hat{y}) = &-\sum_{s=1}^{S}\sum_{b=1}^{B}\left[y_{s,b}^{\top}\log(\mathrm{cls}_b(\mathbf{w}, i_s)) + \mathcal{L}_{\mathrm{loc}}(\mathbf{w})\right] \\ &-\sum_{t=1}^{T}\sum_{b=1}^{B}\Big\{\hat{o}_t\Big[\hat{y}_{t,b}^{\top}\log(\mathrm{cls}_b(\mathbf{w}, i_t)) \\ &\qquad + \mathcal{L}_{\mathrm{loc}}(\mathbf{w}) + k\hat{y}_{t,b}\Big] + \lambda\hat{o}_t\Big\} \\ \text{s.t. } &\hat{y}_{t,b} \in \{0, 1\} \quad \forall t, b \\ &\hat{o}_t \in \{0, 1\} \quad \forall t \\ &k, \lambda > 0 \end{aligned} \tag{8}$$

where $\hat{y}$ and $\hat{o}_t$ represent the pseudo-labels of the proposal level and image level, respectively. $\hat{y}$ denotes the pseudo category labels of proposals, and assigning it as 1 leads to treating this pseudo-label as target ground truth and 0 as background otherwise. $\hat{o}_t$ indicates whether this image will participate in training, and assigning it as 0 denotes unselecting this image in detector training. $k\hat{y}_{t,b}$ and $\lambda\hat{o}_t$ are the regularization term to prevent the trivial solution. k is used to control the number of treating the proposals as background. λ is used to control the number of unselected images. For simplicity, we use $\mathcal{L}_{\mathrm{loc}}(\mathbf{w})$ to indicate the regression term. Note that i_s in our method represents the images of transition domain. We further randomly sample 1:1 with the number of target domains to conduct balanced sampling in each minibatch.

The minimization of revised self-training loss is taken by the alternative block coordinate descent algorithm.

1) Fix $\mathbf{w}$ (previous stage/step) and minimize the objective to generate $\hat{y}_{t,b}$ and $\hat{o}_t$.
2) Fix $\hat{y}_{t,b}$ and $\hat{o}_t$, and then minimize the loss to update $\mathbf{w}$.

Step 1) and step 2) are alternatively repeated. Step 1) selects the most confident proposals and images as pseudo-labels of the SAR domain, and step 2) updates the detector with the pseudo-labels selected in step 1).

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Step 2) can be solved by stochastic gradient descent (SGD) or Adam algorithm, but for step 1), the pseudo-labels are discrete variables and solving it needs nonlinear integer programming. We can rewrite pseudo-labels' generation as follows:

$$\begin{aligned}\min -\sum_{t=1}^{T}\sum_{b=1}^{B}\Big\{\hat{o}_t\Big[\hat{y}_{t,b}^{\top}\log(\mathrm{cls}_b(\mathbf{w},i_t)) \\ +\mathcal{L}_{\mathrm{loc}}(w)+k\hat{y}_{t,b}\Big]+\lambda\hat{o}_t\Big\} \\ \text{s.t. } \hat{y}_{t,b}\in\{0,1\}\quad\forall t,b \\ \hat{o}_t\in\{0,1\}\quad\forall t \\ k,\lambda>0. \end{aligned} \tag{9}$$

Since $\hat{y}_{t,b}$ and $\hat{o}_t$ are required to be $\{0,1\}$, the $\hat{y}_{t,b}$ configuration can be optimized by the solver

$$\hat{y}_{t,b}=\begin{cases}1, & \mathrm{cls}_b(\mathbf{w})\geq\exp(-k)\\ 0, & \text{otherwise}\end{cases} \tag{10}$$

where $\exp(-k)$ is the threshold of probability confidence.

$\hat{o}_t$ can be optimized as follows:

$$\hat{o}_t=\begin{cases}1, & \text{if}\quad \mathrm{cls}_b(\mathbf{w})\geq\exp(-k)\\ & \quad\dfrac{1}{M}\displaystyle\sum_{m=1}^{M}\mathrm{cls}_m(\mathbf{w})\geq\exp(-\lambda)\\ 0, & \text{otherwise}\end{cases} \tag{11}$$

where $(1/M)\sum_{m=1}^{M}\mathrm{cls}_m(\mathbf{w})$ represents the average confidence score of all selected proposal bounding boxes inside an image.

Besides that, we also use label smooth to reduce the influence of noisy labels on detector training

$$\hat{y}'=\hat{y}(1-\alpha)+\alpha/c \tag{12}$$

where α is the pseudo-label smooth factor and indicates the category (ship or background). In step 1), assigning $\alpha=0$ leads to ignoring this smooth term, and in step 2), $\alpha\in(0,1)$.

Then, the formulation can be rewritten as

$$\begin{aligned}\min -\sum_{t=1}^{T}\sum_{b=1}^{B}\Big\{\hat{o}_t\Big[\hat{y}'^{\top}_{t,b}\log(\mathrm{cls}_b(\mathbf{w},i_t)) \\ +\mathcal{L}_{\mathrm{loc}}(w)+k\hat{y}_{t,b}\Big]+\lambda\hat{o}_t\Big\} \\ \text{s.t. } \hat{y}_{t,b}\in\{0,1\}\quad\forall t,b \\ \hat{o}_t\in\{0,1\}\quad\forall t \\ k,\lambda>0. \end{aligned} \tag{13}$$

Note that RPN loss only participates in step 2) and does not participate in the pseudo-label generation process of step 1). Therefore, we present the RST loss of the faster R-CNN for the ROI head for simplicity. More details about RPN loss and the form of RPN loss in semisupervised self-training can be found in the faster R-CNN [6] and STAC [42].

a) Determination of k in RST: k is used to select high-quality pseudo-labels on the proposal level. By setting a threshold $\exp(-k)$, those less confident proposal bounding boxes can be filtered out. $\exp(-k)\in(0,1)$ is a threshold set based on the human experience.

Algorithm 1 Determination of λ in RST

Input: Target SAR training images with pseudo-labels $\boldsymbol{L}=\{\boldsymbol{x_i},\boldsymbol{b_{i,j}},\boldsymbol{c_{i,j}}\}_{i=1}^{T}$, where $\boldsymbol{b_{i,j}}$ and $\boldsymbol{c_{i,j}}$ indicate the proposal box of the $\boldsymbol{ith}$ image and corresponding category probability; confidence threshold of proposal level $\boldsymbol{\exp(-k)}$; image selection portion $\boldsymbol{p}$.

Output: $\boldsymbol{\lambda}$

1: Initialize $\boldsymbol{L_{ins}}$ as an empty set;
2: if $\boldsymbol{c_{i,j}\geq\exp(-k)}$, select $(\boldsymbol{x_i},\boldsymbol{b_{i,j}},\boldsymbol{c_{i,j}})$and add into $\boldsymbol{L_{ins}}$;
3: Calculate $\boldsymbol{score_i=\frac{1}{m_i}\sum_{j=1}^{m_i}c_{i,j}}$. for each image $\boldsymbol{x_i}$ containing $\boldsymbol{m_i}$ bounding boxes in $\boldsymbol{L_{ins}}$;.
4: O=**sort**($\boldsymbol{score_i}$, order=descending);
5: $\boldsymbol{index=length(O)\times p}$;
6: $\boldsymbol{\lambda=-\log(O[index])}$;
7: **return** $\boldsymbol{\lambda}$.

b) Determination of λ in RST: λ is used to select high-quality pseudo-labels on image level. By setting a threshold $\exp(-\lambda)$, those less average confident images can be filtered out. However, the average confidence threshold is difficult to set. To solve this problem, in each round, we design a strategy for determining λ so that we can increase the number of selected images from easy to hard.

We first calculate the average confidence score of all the images from the SAR domain and then sort them according to their average score from high to low. The value of $\exp(-\lambda)$ is set equal to the average score ranked at $(p\times T\times M)$, where $p\in[0,1]$ represents the proportion of selected images. Therefore, $p\times 100\%$ most confident SAR images with pseudo-labels are selected on image level for detector training. We give this pseudo-label selection strategy on image level in Algorithm 1. In particular, p is started from 0.5 and added 0.2 in each additional round.

Besides that, noticing the similarity in problem settings between UDA and semisupervised object detection, RST can also be used for semisupervised object detection tasks and only needs to replace the data of transition domain in UDA with labeled data in semisupervised object detection.

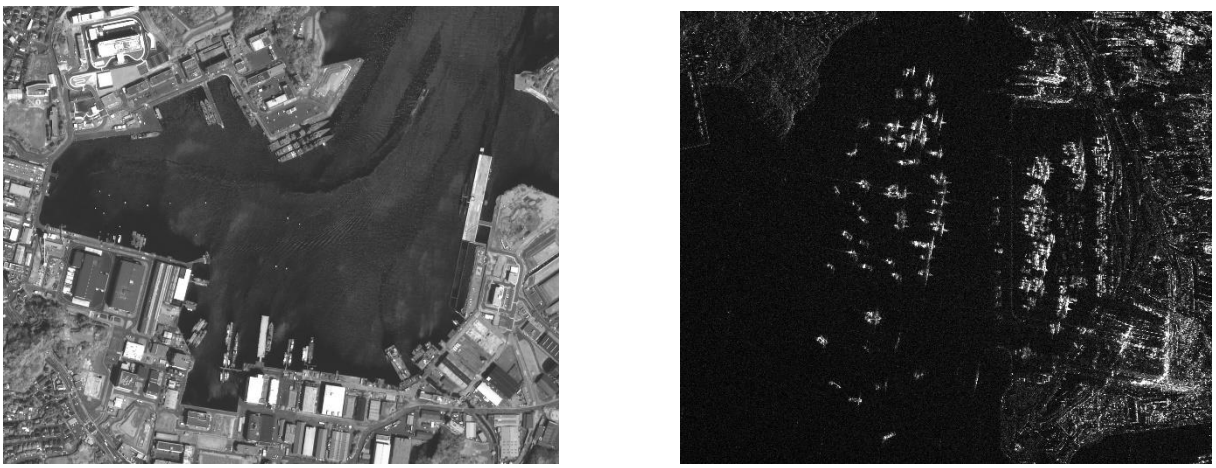
III. Experimental Results and Analysis

A. Experimental Data Description and Implement Details

This section evaluates our method based on domain adaptation from the optical to SAR remote sensing ship detection dataset. The SAR ship detection dataset is public AIR-SARship-1.0 [45], and the optical ship detection dataset is collected by the Chinese GF-2 satellite (GF2ship detection dataset). Fig. 6 shows two images of GF2ship and AIR-SARship-1.0.

1) GF2ship: The GF2ship dataset includes five optical images with the size of $29\,200\times 27\,620$ and 37 optical images with the size of 4000×4000, which is collected from the Chinese GF-2 satellite. The resolution of the GF2ship dataset is 0.8 m. We use the GF2ship dataset as the source domain and crop the original optical images to 1280 subimages with the size of 512×512. The GF2ship dataset contains 2110 ships.

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(a) (b)

Fig. 6. Examples of the GF2 ship dataset and AIR-SARship-1.0 dataset. (a) Example of the GF2 ship dataset. (b) Example of the AIR-SARship-1.0 dataset.

2) AIR-SARship-1.0: The publicly released AIR-SARship-1.0 dataset consists of 31 SAR images of large scenes with 879 ships and includes images with the resolution of 1 and 3 m. The dataset is divided into a training set and a test set, where the training set contains 374 subimages with the size of 512×512, which is cropped from the original 21 SAR images with the size of 3000×3000, and the test set contains the remaining ten original images with the size of 3000×3000. In our experiment, the AIR-SARship-1.0 dataset is used to be the target domain.

In our experiment, the optical GF2ship dataset is used to be the source domain, where the rectangle bounding boxes and rotatable bounding boxes of ships can be accessed. The SAR AIR-SARship-1.0 dataset is used to be the target domain, where only the images without labels can be accessed. The target domain data are divided into two parts following the official division, one part is the training set, which participates in training for UDA with all data of the optical GF2ship dataset; the other part is the test set, which is used to test the domain adaptative model. During the test, the original ten test images of the AIR-SARship-1.0 dataset were cropped into 512×512 subimages through a sliding window repeatedly. The ship detector detects all the subimages and restores the detection results to the original SAR images.

As for the proposed method, α and β of (5) are set as 2 and 10, respectively; λ of (6) is set as 0.15; the learning rate of feature level and prediction level is 0.001, and the decay is set as 0.1. We set the threshold $\exp(-k)$ of (10) as 0.8. Our method is implemented with Pytorch.

B. Evaluation Metrics

The precision, recall, and F1-score are used to evaluate different detector performances quantitatively, and the calculation is formulated as follows.

1) Precision and Recall: Precision and recall are two widely used criteria for evaluating the performance of detectors in SAR images, which are formulated as

$$\text{Precision} = \frac{N_{\text{td}}}{N_d} \tag{14}$$

$$\text{Recall} = \frac{N_{\text{td}}}{N_{\text{gt}}} \tag{15}$$

where N_{td} denotes the number of correctly detected ship targets, N_d denotes the number of detected ships, and N_{gt} represents the ground truth. The precision and the recall measure the false alarm rate and missed detection rate, respectively.

2) F1-Score: F1-score is the harmonic mean between precision and recall and is formulated as

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Precision}} \tag{16}$$

where F1-score comprehensively evaluates the performance of the detector.

C. Performance Comparison

In this section, we compare our method with the traditional unsupervised Gaussian CFAR and two other famous UDA detection methods in computer vision to demonstrate that our approach achieves superior SAR ship detection performance. Table I and Fig. 7 present the comparison results. In Table I and Fig. 7, Gaussian CFAR denotes the conventional unsupervised detection method in [5]. DAF and hierarchical transferability calibration network (HTCN) represent the UDA detection method in [24] and [28], respectively. Faster RCNN-supervised denotes faster R-CNN method with fully supervised SAR images. Faster RCNN-supervised with transition domain further adds transition images in the training process based on the faster RCNN-supervised method, which is the oracle upper bound result. We also compared with other supervised object detection methods in Table I: SSD [36], RefineDet [37], and FPN [38].

Table I gives the detection results of different methods with precision, recall, and F1-score on ten test SAR images in the AIR-SARship-1.0 dataset for quantitative analysis. As can be

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TABLE I
COMPARISON OF THE OVERALL EVALUATION OF VARIOUS METHODS

Methods	Precision	Recall	F1-score
Gaussian CFAR	0.4014	0.9040	0.5559
DAF	0.4792	0.1840	0.2659
HTCN	0.6061	0.4800	0.5357
Proposed method	0.8293	0.8080	**0.8185**
Faster RCNN-supervised	0.7483	0.8800	0.8122
Faster RCNN-supervised with transition domain (Oracle)	0.8110	0.8400	0.8253
SSD-supervised	0.7966	0.7600	0.7779
FPN-supervised	0.8000	0.8480	0.8233
RefineDet-supervised	0.8121	0.8550	0.8329

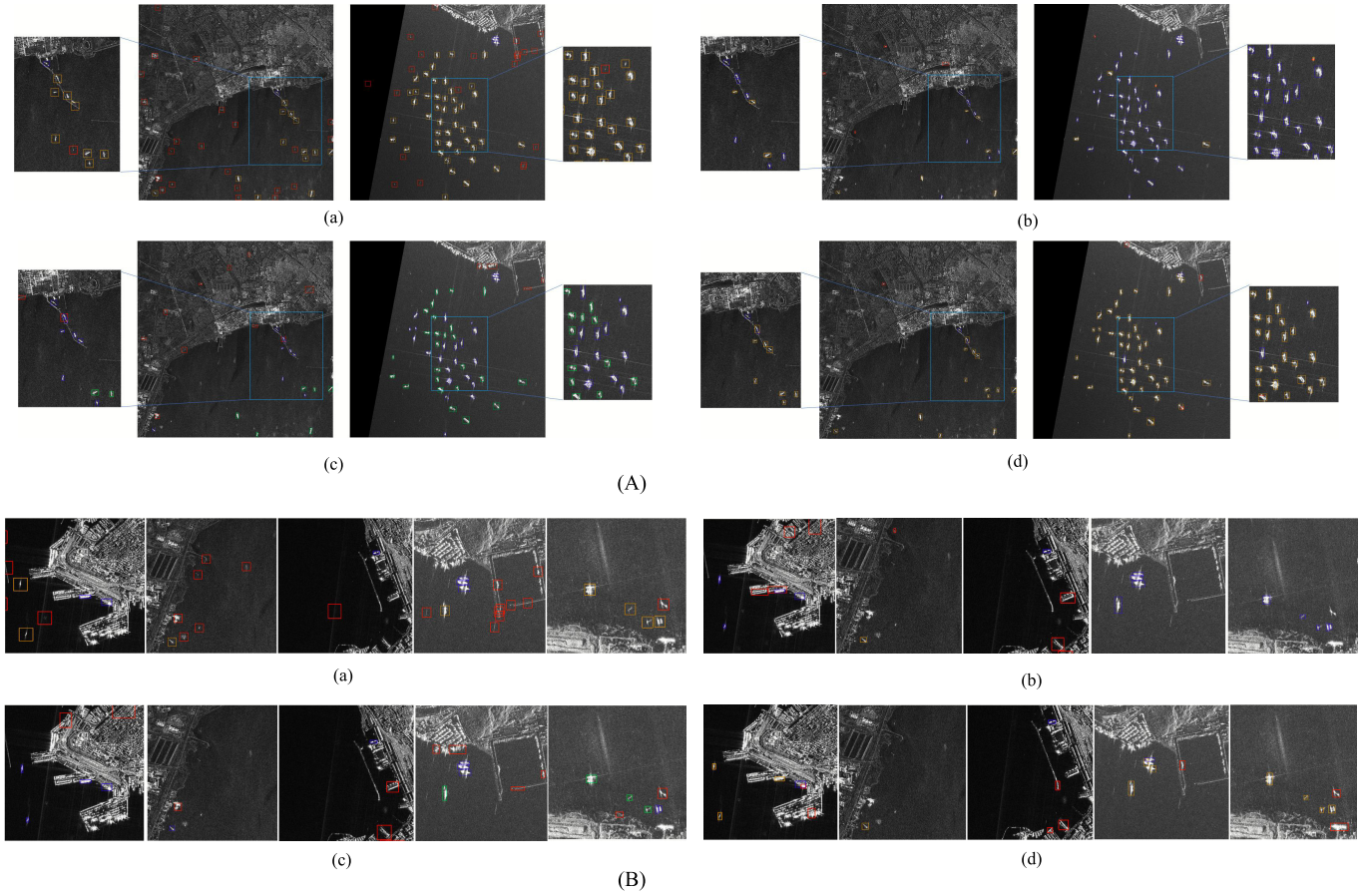


Fig. 7. Visual detection results of the proposed method and comparison methods. (A) (a)–(d) Detection results of large scenes. (B) (a)–(d) Detection results of inshore and port; each image is cropped from the original large scene SAR image. (a)–(d) Correspond to the results of the four methods Gaussian CFAR, DAF, HTCN and the proposed method, respectively. Rectangles with yellow color mark the ships that are correctly detected, rectangles with blue color mark the missed ships, and rectangles with red color mark the falsely detected results.

seen from Table I, the precision and comprehensive criterion F1-score of Gaussian CFAR are very low. Because Gaussian CFAR is a conventional unsupervised detection method and its detection performance would be reduced in complex scenes. In terms of UDA detection methods, due to the significant difference between the optical domain and the SAR domain, it is not ideal to directly apply the methods in computer vision to the cross-domain detection of optical to SAR. The performance of DAF and HTCN is even worse than the CFAR-based method. The performance of DAF is the worst since it only transfers knowledge at feature level. Compared with DAF, HTCN uses CycleGAN to reduce the pixel level difference and achieve a better result. But HTCN aligns feature distributions at low-, middle-, and high-level feature spaces, limiting the improvement of detection performance. The proposed method achieves the best detection performance among all the methods, which verifies the superiority of the proposed method. It is worth noting that our method

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TABLE II
ABLATION STUDY

Methods	OSGAN	Synthetic	Feature	Prediction	Precision	Recall	F1-score
CycleGAN	-	-	-	-	0.5556	0.4800	0.5150
Ours	✓				0.6149	0.7280	0.6667
	✓	✓			0.6788	0.7280	0.7026
	✓		✓		0.6871	0.7920	0.7358
	✓	✓	✓		0.7143	0.7920	0.7511
	✓	✓		✓	0.7464	0.8160	0.7796
	✓	✓	✓	✓	0.8293	0.8080	**0.8185**
Oracle	-	-	-	-	0.8110	0.8400	0.8253

outperforms the traditional supervised faster R-CNN, and the traditional supervised faster R-CNN can achieve better performance with the help of transition images. Compared with SSD, faster R-CNN has higher F1-score performance. Due to the multiscale pyramid mechanism, FPN based on faster R-CNN achieves better performance. RefineDet has achieved the best performance compared with other supervised methods by introducing the idea of anchor refinement and multiscale feature fusion. But FPN and RefineDet have more complex structure compared with faster R-CNN.

Fig. 7 exhibits some visual detection examples of the proposed method and other comparison methods. These scenes are all contained in the test set. As shown in Fig. 7, though Gaussian CFAR can detect almost all ships correctly, there are a lot of false alarms in the detection results and the method cannot accurately locate the targets (the same size of boxes). Due to the enormous difference between the optical and SAR images, DAF and HTCN have a large amount of missed detections. Compared with DAF, HTCN has fewer missing detections by reducing the appearance difference. The proposed method is beyond all comparison methods and only has fewer false alarms and missed detections at the junction of land and sea. We further show the comparison results with other methods in complex scenarios in Fig. 7(B). Our method has fewest false alarms and missed detections. Besides that, our method can detect some ships moored in port [first image in (d)].

D. Model Analysis

1) Ablation Study: To comprehensively investigate the effectiveness of different levels on detection performance, we conduct the ablation study by evaluating variants of the proposed method. The ablation study results are shown in Table II, where OSGAN, Synthetic, Feature, and Prediction present whether to use the optical-to-SAR GAN, image synthetic, domain-invariant feature representation, and RST, respectively. We can see that all the proposed modules are designed reasonably and all the modules can improve the performance. Furthermore, transferring knowledge from the pixel, feature, and prediction levels significantly boosts the performance for overall cases. We also compare our OSGAN with the traditional CycleGAN. Compared with CycleGAN, our OSGAN achieves +15.17% F1-score performance, which demonstrates the effectiveness of OSGAN. At the pixel level, image synthetic can further improve the detection performance. Furthermore, precision = 1-FAR, where FAR represents false alarm rate. We can find that our image synthetic for ship targets can significantly reduce the false alarms. Because our special data augmentation reduces the impact from ship wakes and complex land backgrounds on ship data generation. Meanwhile, our method enhances the diversity of ship targets data. The statistical learning theory [46] shows that the generalization of models can be presented by the diversity of training data. Therefore, our method can improve the generalization ability by enhancing the diversity. Compared with feature alignment, RST achieves better results by robustly learning more discriminative features of the SAR domain, but RST depends on the performance of the previous stage detector. If the detector of the previous stage has too poor performance, it is difficult for RST to achieve good results. In addition, due to our pseudo-label selection strategy, RST can effectively reduce false alarms. Self-training assumes that the higher the confidence of the proposal, the higher the probability that the proposal is the object [42], [47], [48], i.e., the confidence of the clutter proposal is relatively low compared with the ship object proposal. Our method selects the images participating in training by calculating the average score of the proposals in an image. The images containing more false alarms (clutter proposals) will have a low average score. Then these images containing more false alarms will be screened out so that the probability of the detection model misjudging the clutter as the target is reduced, which is beneficial to reduce false alarms.

2) Visualization of Pixel Level: The visualized results of optical-to-SAR image generation are presented in Figs. 8 and 9. Fig. 8 shows the generation results of OSGAN and comparison CycleGAN. As can be seen from Fig. 8, the generated transition images of OSGAN are very similar to real SAR images. Furthermore, compared with CycleGAN, the images generated by our OSGAN better retain the edge, structure, and shape information, which demonstrates the rationality of our method. Fig. 9 shows the generation results of original

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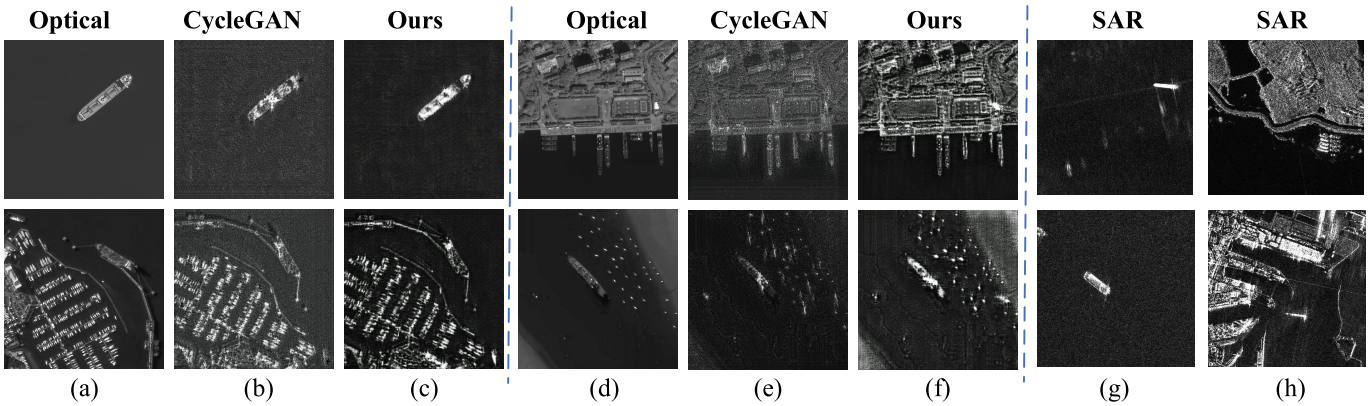


Fig. 8. Visualized generation results of OSGAN and comparison CycleGAN. (a) and (d) Original optical images, (b) and (e) corresponding generation results of CycleGAN, (c) and (f) corresponding generation results of our OSGAN. (g) and (h) Four images of real SAR images in AIR-SARShip-1.0.

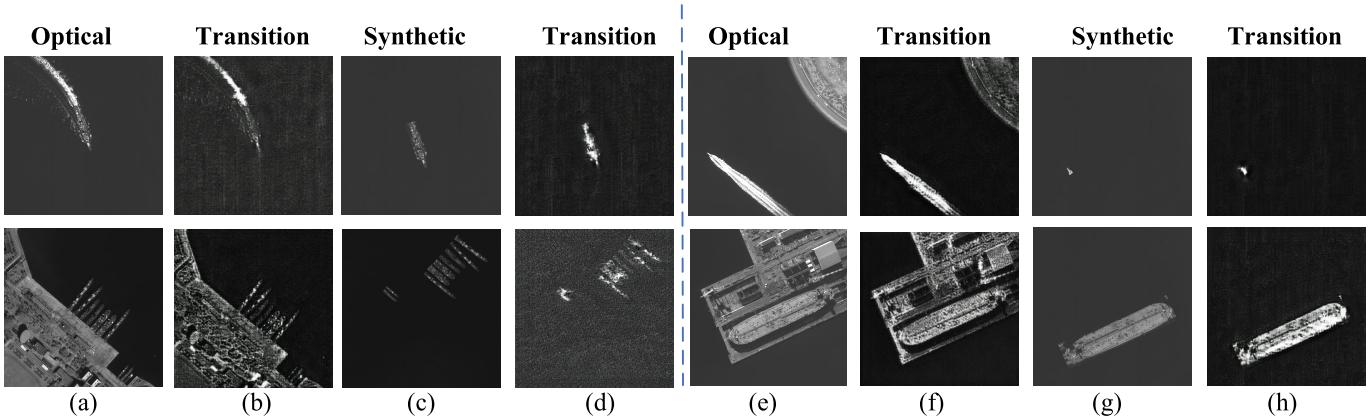


Fig. 9. Visualized generation results of original optical images and synthetic images. (a) and (e) Original optical images, (c) and (g) corresponding synthetic images without wakes and lands, and (b), (f), (d), and (h) corresponding generation results of original optical images and synthetic images.

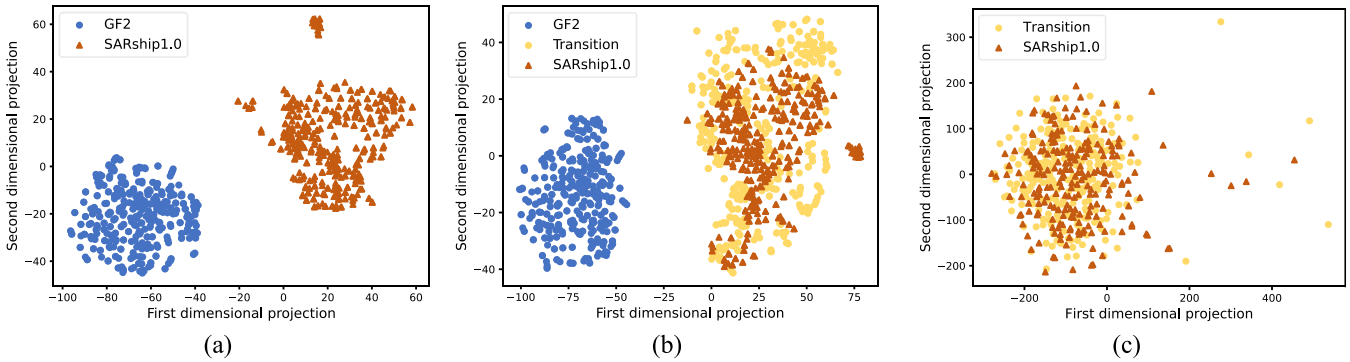


Fig. 10. Visualized t-SNE results of feature distributions, where each blue dot denotes one image feature of the optical GF2ship dataset, each red triangle denotes one image feature of the SARship dataset, and each orange blot denotes one image feature of the transition domain. All the features are extracted by the last convolutions of VGG16. (a) Original results of the optical domain and SAR domain. (b) Result after pixel-level adaptation. (c) Result after feature-level adaptation.

optical images and synthetic images. Compared with original optical images, synthetic images only retain the ship target without wakes and complex lands, which can be seen as a special data augmentation for the ship target. As shown in Fig. 9, synthetic images generate better ship targets. We can conclude that the synthetic images can better focus on the characteristics of the ship targets and avoid the influence of the wakes and complex land background on the generation of ships.

To further verify the effectiveness of our optical-to-SAR image generation for pixel domain adaptation, we give the feature distribution plot of the optical domain, transition domain, and SAR domain via t-SNE. Fig. 10(a) and (b) shows

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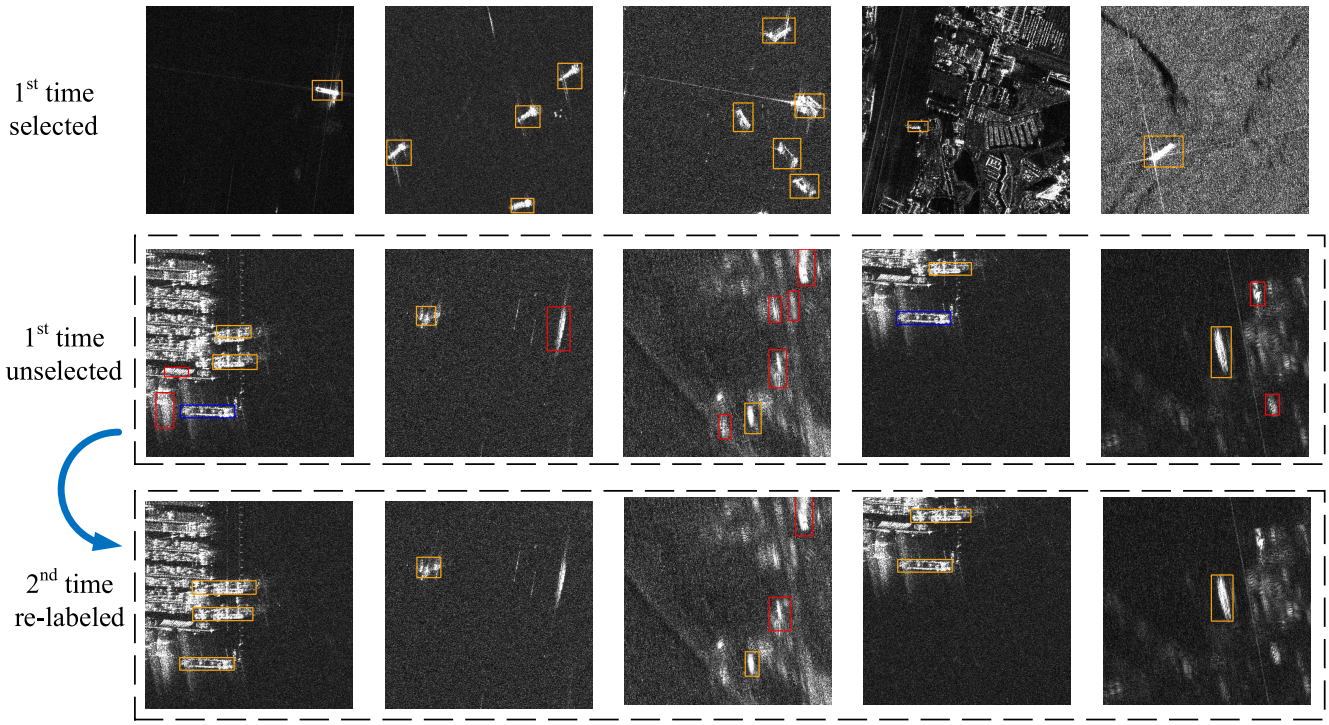


Fig. 11. Pseudo-label generation and regeneration results for SAR images of AIR-SARship-1.0, where rectangles with yellow color mark the ships that are correctly detected, rectangles with blue color mark the missed ships, and rectangles with red color mark the falsely detected results. Some easy images (high average image score) selected by the detector are listed in the first row. Some difficult images (low average image score) not selected by the detector are listed in the second row. The third row lists difficult images corresponding to the second row that are regenerated by the detector after updating the parameters of the model.

the results of pixel adaptation from the GF2ship dataset to AIR-SARship-1.0 by mapping the corresponding features to the 2-D space. As shown in Fig. 10, the SAR domain has an extensive distribution gap with the optical domain due to the prominent difference between SAR images and optical images. Fig. 10(b) presents the result after pixel-level adaptation. Our generated transition domain reduces the domain gap significantly and nearly cannot be distinguished from the real SAR domain, demonstrating the superiority of our method.

3) Analysis of Feature Level: We also give the visualized t-SNE after feature-level adaptation in Fig. 10(c), where transition features and SAR features are extracted by the same extractor after feature level alignment. Compared with the distribution plot in Fig. 10(b), the distribution of real SAR features after feature alignment is further aligned and closer to the distribution of transition features.

To further explore how to transfer in deep detector from optical to SAR, Table III evaluates variants of the feature alignment at different levels, where low level, middle level, and high level indicate aligning feature at low level (third convolutional blocks of feature extractor), aligning feature at middle level (forth convolutional blocks of feature extractor), and aligning feature at high level (fifth convolutional blocks of feature extractor), respectively. Table III shows that the detector achieves the best F1-score performance by aligning the feature distribution at high-level semantic feature space and aligning the feature distribution at low-level feature space is worst. These results validate our hypothesis.

TABLE III
RESULTS OF ALIGNMENT IN DIFFERENT LEVELS

level	Precision	Recall	F1-score
Low-level	0.6364	0.8560	0.7300
Middle-level	0.6771	0.8240	0.7433
High-level	0.7143	0.7920	**0.7511**

4) Analysis of Prediction Level: RST can robustly learn knowledge from pseudo-labels and update the detector to regenerate labels for unselected images. To qualitatively analyze this process, we give pseudo-label generation and regeneration results for SAR images of AIR-SARship-1.0 in Fig. 11. The pseudo-labels generated by the initial detector have some erroneous knowledge and some correct knowledge. The images that contain more erroneous pseudo-labels will hurt the detector training. The first row in Fig. 11 shows that easy images (high image average score) selected by the detector for the first time and the second row shows the unselected difficult images (low image average score). As mitigation, our method can screen out most images with low-quality pseudo-labels, making the detector training robust. The third row of Fig. 11 shows difficult images corresponding to the second row that are regenerated by the detector after updating the parameters of the model. The update detector can iteratively generate pseudo-labels with higher precision/recall, which improves the quality of pseudo-labels. As mentioned in Section II-C,

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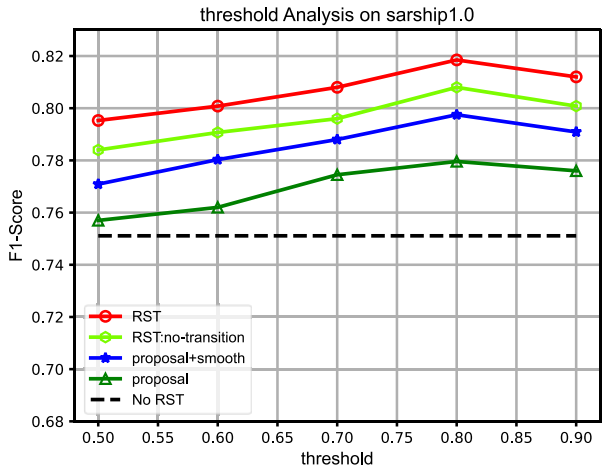


Fig. 12. Ship detection performance with different variants and threshold, and F1-score of our RST reaches the peak when the threshold is 0.8.

we set a selection portion p to control this process, where p represents the proportion of selected images to all training SAR images. Our method has at most four rounds/iterations. p is set as 0.5, 0.7, and 0.9, 1 in each round/iteration. Specifically, in our AIR-SARship-1.0 experiments, 51 SAR images (about 13% of the total training SAR images) cannot generate pseudo-labels since our model cannot detect any ships of these images. Therefore, our method has only three rounds/iterations in our experiment. This process costs more training time in the training phase but does not add extra testing cost.

To explore the effectiveness of our RST, we also perform RST with different variants and thresholds. The results are plotted in Fig. 12, where "No RST," "proposal," "proposal+smooth," "RST: no-transition," and "RST" represent the detector trained without pseudo-labels, the detector trained with only proposal-level pseudo-labels, the detector using label smooth based on "proposal," the RST detector trained without transition images, and our RST detector, respectively. In terms of threshold, as shown in Fig. 12, higher thresholds achieve better detection performance than lower thresholds because the initial teacher model (feature aligned detector) has a relatively low precision (more false alarms). The F1-score of our RST reaches the peak when the threshold is 0.8. The results of different variants demonstrate that all the proposed strategies of RST are designed reasonably and all the strategies can improve the detection performance. Our variants of RST all outperform the method without using pseudo-labels because the detector can directly learn more discriminative knowledge from the SAR images. Compared with the invariants of RST, the full version RST achieves the best F1-score detection performance. The compelling results significantly verify that the proposed RST can robustly learn discriminative knowledge from SAR images by iteratively producing pseudo-labels and retraining the detector with robust strategies.

E. Discussion

1) What Knowledge Has Been Transferred: In traditional transfer learning, the model parameters of the source domain are transferred to the target domain, and then the labeled target domain data are used for fine-tuning. The target domain training data under UDA setting is unlabeled, and it is impossible to perform supervised learning directly. Therefore, it needs to transfer the similarity of the two domains to make the labeled optical data guide the learning of unlabeled SAR data. As pointed by the survey article [49], most UDA methods transfer the commonality/similarity knowledge between the source domain and the target domain to find a good feature representation for the target domain that reduces the error of classification and regression models. Same as most UDA methods, the transferred knowledge of our methods is the similarity between the optical domain and SAR domain and aims to find a good feature representation for the ship detection task of the SAR domain. Then we will discuss and illustrate the similarity between our optical and SAR domains and further visualize the "good feature representation."

For the ship detection task, ships of the optical domain and the SAR domain contain similar characteristics and some different characteristics. Similar characteristics may help improve the ship detection performance of the SAR domain. Fig. 13 shows some examples of similar characteristics, where (a)–(d) shows the size distribution, aspect ratio distribution, shapes, and arrangements of ships in the optical domain (GF2ship dataset) and SAR domain (AIR-SARship-1.0 dataset), respectively. As illustrated in Fig. 13(a), the ship sizes of the optical domain contain all the ship sizes of the SAR domain, and the optical domain has more ships. As can be seen from Fig. 13(b), the ship aspect ratios of optical and SAR domains are similar. Fig. 13(c) and (d) shows the similarity examples of ship shapes and arrangements. The above gives some simple examples that are obvious and easy to be understood. We believe that there are still other implicit similarities contained in the two domains, which can be learned by the deep networks automatically.

As mentioned in the previous paragraph, our UDA transfers the similarity between the optical and the SAR domains to find a good feature representation for SAR domain. We further visualize the "good feature representation" and explore the similarity of deep feature representation. Fig. 14 gives the visualization of deep feature representation extracted by the last layer of feature extractor (VGG16 in our faster R-CNN), where (a)–(c) represent the feature maps of optical images extracted by the supervised faster R-CNN trained on the GF2ship dataset, feature maps of SAR images directly extracted by the supervised faster R-CNN trained on the GF2ship dataset without adaptation, and feature maps of SAR images extracted by our UDA faster R-CNN model, respectively. These showed images do not participate in the training phase. As illustrated in Fig. 14, supervised faster R-CNN trained on the GF2ship dataset can extract "good features" (enhance the ship features and suppress other background features) for optical ships. But when this supervised faster R-CNN trained on the GF2ship dataset directly extracts the features for the SAR images of AIR-SARship-1.0, these features are too "bad" to be detected by the detector model. Our UDA faster R-CNN model can extract "good" features for the SAR images (similar to the optical features) since the similarity knowledge is transferred.

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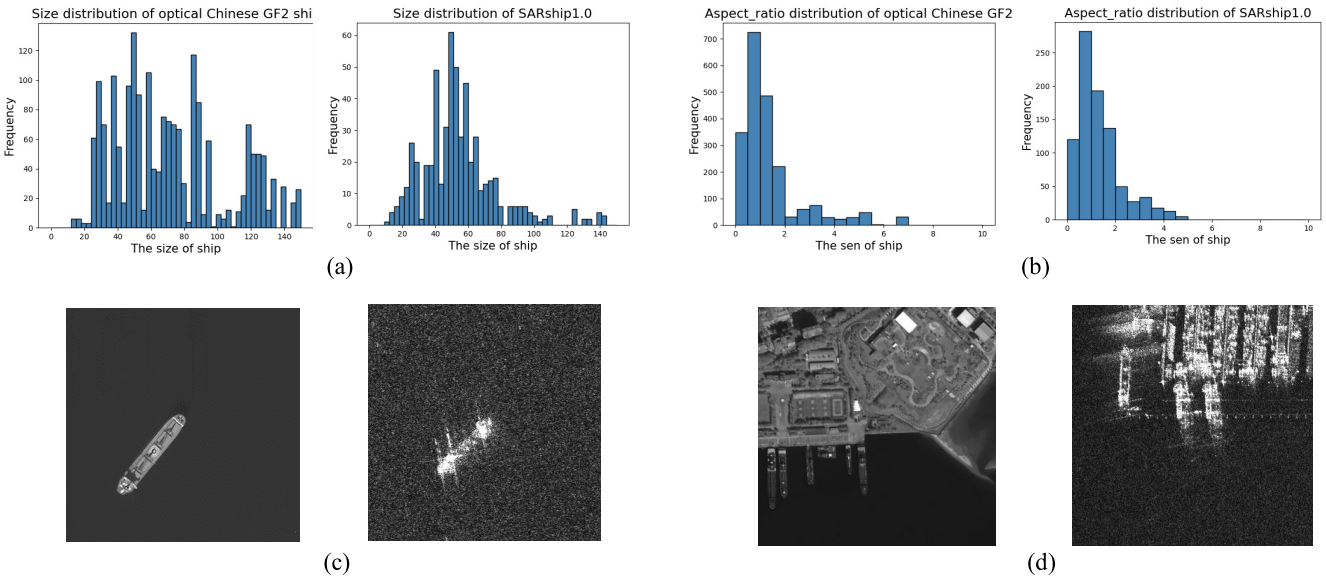


Fig. 13. Ship similarity comparison between the optical domain and SAR domain. (a)–(d) (Left) Example of the optical GF2ship dataset and (Right) Example of the SAR AIR-SARship-1.0 dataset. (a) Size distribution of two domains: the size is equal to the product of ship length and width. (b) Aspect ratio distribution of two domains: the aspect ratio is equal to the ratio of ship length to width. (c) Shape examples of ship targets of two domains. (d) Arrangement examples of ship targets of two domains.

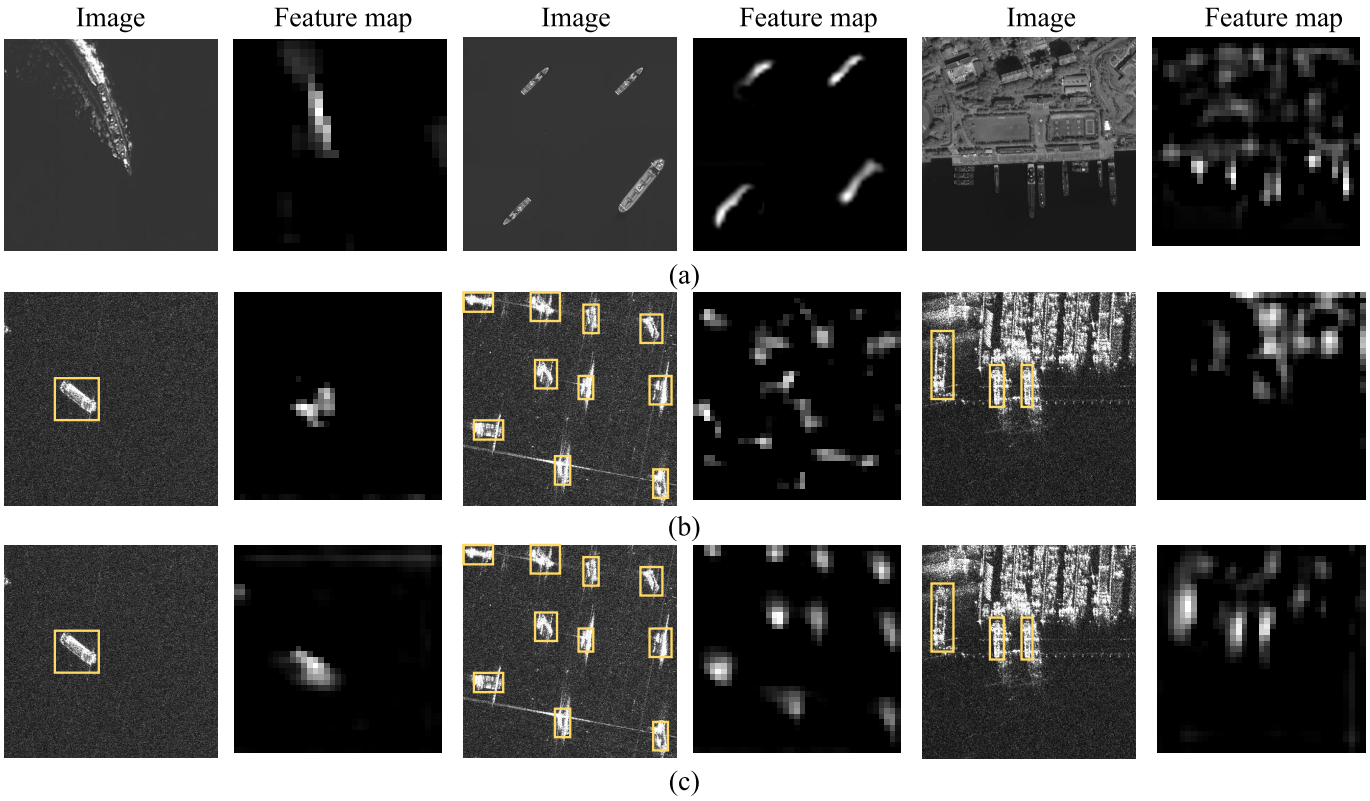


Fig. 14. Visualization of deep feature representation. (a) Optical GF2ship images and corresponding feature maps extracted by the supervised faster R-CNN trained on the GF2ship dataset. (b) SAR images and the corresponding feature maps directly extracted by the supervised faster R-CNN trained on the GF2ship dataset without adaptation. (c) SAR images and the corresponding feature maps extracted by our UDA faster R-CNN model. SAR images in (b) and (c) are same, and yellow rectangles represent ship targets in SAR images.

2) Why Our Method Contributes to Detection Task: Similar characteristics of the two domains may help improve the performance for the ship detection task of the SAR domain. But the optical domain and the SAR domain also contain some different characteristics. For example, due to different imaging mechanisms, the appearance of optical images and

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SAR images are greatly different. In addition, optical ships are susceptible to wakes, while SAR ships are susceptible to speckle noise and there exists the problem of missing structure and so on. These dissimilarities will affect the knowledge transfer from the optical domain to the SAR domain. Therefore, for UDA methods, the first thing we need to do is to reduce the differences between two domains and transfer the similarities between the two domains. The goal of pixel level and feature level of our method is to reduce the difference between the domains and to transfer the similarity between two domains so that the labeled optical data can guide the learning of unlabeled SAR domain training data. Chen *et al.* [24] provide a probability view to explain why this contributes to the detection task of the unlabeled target domain. We give the probability view in the following. Some symbols and forms have been modified to suit our optical to SAR ship detection tasks.

The ship detection problem can be viewed as learning the posterior $P(B|F(I))$, where I is the input image, $F(I)$ is the corresponding feature representation, and B is the bounding box of a ship object.

Let us denote the joint distribution of training samples for ship detection as $P(B, F(I))$, and use $P_O(B, F(I))$ and $P_S(B, F(I))$ to denote the optical domain joint distribution and the SAR domain joint distribution, respectively. Note that here we use $P_S(B, F(I))$ to analyze the domain difference, although the bounding boxes are unknown during training. When there is a domain difference, $P_O(B, F(I)) \neq P_S(B, F(I))$.

Using Bayes' formula, the joint distribution can be decomposed as

$$P(B, F(I)) = P(B|F(I))P(F(I)). \tag{17}$$

For an ideal ship detector, it can detect ship targets in an image, whether it is an optical or SAR image, i.e., the conditional probability $P(B|F(I))$ is the same for the two domains. The domain distribution difference is caused by the difference in the marginal distribution $P(F(I))$. In the faster R-CNN model, the image representation $F(I)$ is actually the feature map output of VGG16. Therefore, to handle the unlabeled SAR ship detection problem, we should enforce the distribution of image feature representation from two domains to be the same (i.e., $P_O(F(I)) = P_S(F(I))$). Then we can obtain a detector that is beneficial to the unlabeled SAR domain. In our method, I presents the pixel level and $F(I)$ presents the feature level. Ideally, one can perform domain adaptation on either pixel level or feature level. However, due to the large difference between the optical and SAR domains, it is hard to align the marginal distribution individually. We align the marginal distribution separately at the pixel level and feature level. We propose the optical-to-SAR image generation and use GANs loss to learn the domain-invariant features.

Though feature level can learn domain-invariant features to align the distributions and improve the detector's performance, it will lose some domain-specific features [50] that may be useful for the detection of the SAR domain. RST of prediction level can directly learn the discriminative features of SAR images through pseudo-labels, which is helpful for detection tasks. Our prediction level is to further use the model learned by mining the similarity in the first two levels to predict high-quality pseudo-labels.

3) Resolution Requirement Between SAR and Optical Data: The resolution of the optical GF2ship dataset is 0.8 m, and the resolution of SAR AIR-SARship-1.0 is 1 and 3 m. Our method can achieve superior performance when the resolution between the optical and SAR domains does not have a very large difference. But when the resolution between the optical and SAR domains has a huge difference, such as the resolution of the SAR domain is 10 m or larger, our method may have a performance drop. If the resolution difference is too large, the similarity between two domains is difficult to transfer, and there may even be negative transfer [49]. If the resolution of SAR domain is too low, the ship size will change significantly, making the two domains more dissimilar. In addition, the SAR image quality will be significantly reduced, due to the ship shape and some other features being distorted.

IV. Conclusion

In this article, we propose an UDA based on progressive transfer for ship detection, which can gradually transfer the knowledge between the optical and the SAR domains from pixel level, feature level, and prediction level. At the pixel level, the synthesis imaged combined with ship targets and pure ocean background alleviates the wakes of ships and complex land background on ship target generation, generating diversity and high-quality ship targets. Besides that, the generator built with skip connection can preserve low-level texture, structure, and shape information to generate more satisfied SAR images. We explore where there is more effective to align the feature distributions between the optical domain and the SAR domain at the feature level. The domain-invariant features learned at the high-level feature space can achieve a better result. At the prediction level, the proposed RST can robustly learn discriminative knowledge from SAR images by iteratively producing pseudo-labels and retraining the detector with robust strategies. Specially, RST is formulated as a loss minimization problem for object detection. The compelling results based on domain adaptation from the optical to SAR ship detection dataset significantly demonstrate the effectiveness of the proposed UDA detection method.

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